

YOUSOF RAHIMI

M.Sc. in Artificial Intelligence – AI Research Engineer

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RESEARCH INTERESTS

Approximate Message Passing, Graph Neural Networks, Kalman filtering, AI-driven drug design, metabolic engineering, material informatics, AI agentic systems.

EDUCATION

M.Sc. Artificial Intelligence — Amirkabir Univ. of Technology (Tehran Polytechnic), 2024–2026 (GPA 4.0/4.0)

Thesis: Message Passing and Graph Neural Networks for High-Dimensional Inference.

B.Sc. Computer Engineering — University of Tehran, 2019–2023 (Top 3%)

PUBLICATIONS (SELECTED)

- GAMP-GNN: Message Passing Meets GNNs for Model-Based Compressed Sensing — ICML 2026 (under review)
- DiffMolecule: Guided Diffusion over Graphs for De Novo Drug Design — JCIIM (in press)
- Data-Driven Flux Prediction in Genome-Scale Metabolic Models — Metabolic Engineering (2025)
- CrystalFormer: Bond-Aware Transformer for Inorganic Property Prediction — npj Comput. Mater. (2025)
- Invariant EKF with Learned Noise Models for Multi-Sensor Fusion — IEEE TAES short (2024)

RESEARCH & WORK EXPERIENCE

Research Assistant — Sparse Inference Lab, 2024–Present

AMP-based iterative algorithms and GNN message-passing layers for compressed sensing; PyTorch prototypes.

ML Engineer (part-time) — Computational Chemistry Startup, 2025–Present

Owned ADMET property prediction; 2x virtual screening throughput; retrosynthesis prototype.

Teaching Assistant — Machine Learning, 2024–2025

Designed Bayesian-methods labs and 10+ Jupyter notebooks for 90+ students; TA excellence award.

TECHNICAL SKILLS

Languages: Python, Julia, C++, SQL

ML / DL: PyTorch, PyG, JAX, TensorFlow, Scikit-learn

Sci. tools: RDKit, OpenMM, GROMACS, COBRApy, ASE/Pymatgen

Theory: Probability, Optimization & Duality, Filtering & Control, Graph/Spectral theory